

Updates & Notes

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Inverse Dynamics $\rightarrow$ Forward Kinematics Pipeline

Idea: Learn control representations instead

1. Train ID model: $(I_t, I_{t+1}) \rightarrow (v_L, v_R)$ [supervised]
2. Apply FK: $(v_L, v_R) \rightarrow (\dot{x}_{body}, \dot{\theta}_{body})$ [deterministic]
3. Integrate for pose predictions

Does this preserve enough information?

- ▶ Invertibility $\rightarrow$ bijective
- ▶ bijective $\rightarrow$ *no* information loss
 - ▶ inputs can be reconstructed from outputs

Bijection Analysis (differential drive example)

(in the body frame)

$$\dot{x}_{body} = v = \frac{v_L + v_R}{2}$$

$$\dot{y}_{body} = 0$$

$$\dot{\theta} = \omega = \frac{v_R - v_L}{L}$$

Bijection Analysis: Bicycle Model

Bicycle model dynamics (body frame):

$$\dot{x}_{body} = v$$

$$\dot{y}_{body} = 0$$

$$\dot{\theta} = \frac{v \tan(\delta)}{L}$$

where v is velocity, δ is steering angle, L is wheelbase.

Controls: (v, δ) or $(\dot{v}, \delta)$ depending on whether velocity is controlled directly

Inverse dynamics (body frame):

$$v = \dot{x}_{body}$$

$$\delta = \arctan \left(\frac{L\dot{\theta}}{\dot{x}_{body}} \right)$$

Bijectivity: $(\dot{x}_{body}, \dot{\theta}) \leftrightarrow (v, \delta)$